Linux Fundamentals

PROJECT: INFO EXTRACTOR

1. Script overview:



```
 GNU nano 8.7                                                    TCB-2511.s6.xe103.sh
#!/bin/bash

#1. Identify the system's public IP.
echo "[*] Public IP Found:"
curl -s ifconfig.co

#2. Identify the private IP address assigned to the system's network interface.
echo "[*] Private IP Address Found:"
ifconfig | grep broadcast | awk '{print $2}'

#3. Display the MAC address (masking sensitive portions for security).
echo "[*] MAC Address found:"
echo "$(ifconfig | grep ether | awk '{print $2}'|awk -F ':' '{print $1,$2,$3}' | tr ' ' ':'):xx:xx:xx"

#4. Display the percentage of CPU usage for the top 5 processes.
echo "[*] Top Processes:"
ps -eo "pid,%cpu,comm" --sort=-%cpu | head -n6

#Display memory usage statistics: total and available memory.
echo "[*] Memmory:"
memmory=$(free -h | grep Mem)
echo " Total: $(echo $memmory | awk '{print $2}')"
echo " Available: $(echo $memmory | awk '{print $NF}')"

#6.List active system services with their status.
echo "[*] Active services:"
service --status-all | grep -F '[ + ]'

#7. Locate the Top 10 Largest Files in /home.
echo "[*] Top Files:"
du -ha /home 2>/dev/null | sort -n | tail -n10
```

2. Output after executing the script:



```
┌──(evgeni㉿kali)-[~/ThinkCyber/Project/xe103]
└─$ ./TCB-2511.s6.xe103.sh
[*] Public IP Found:
217.217.112.161
[*] Private IP Address Found:
192.168.50.186
[*] MAC Address found:
74:d4:dd:xx:xx:xx
[*] Top Processes:
    PID %CPU COMMAND
   5473 10.8 firefox-esr
   3175  9.4 gnome-shell
   4862  4.2 Discord
 182282  2.4 Isolated Web Co
 104213  1.3 Isolated Web Co
[*] Memmory:
 Total: 30Gi
 Available: 24Gi
[*] Active services:
 [ + ]  binfmt-support
 [ + ]  cron
 [ + ]  dbus
 [ + ]  gdm3
 [ + ]  networking
 [ ? ]  nvidia-persistenced
 [ + ]  pcscd
 [ + ]  plymouth-log
 [ + ]  procps
 [ + ]  smartmontools
[*] Top Files:
1020K   /home/evgeni/WebstormProjects/functions/node_modules/@opentelemetry/api/build/esnext
1020K   /home/evgeni/WebstormProjects/KapanaLuxuryApartment/functions/node_modules/@firebase/database/dist/index.standalone.js.map
1020K   /home/evgeni/WebstormProjects/KapanaLuxuryApartment/functions/node_modules/@opentelemetry/api/build/esnext
1020K   /home/evgeni/WebstormProjects/KapanaLuxuryApartment/node_modules/@opentelemetry/api/build/esnext
1020K   /home/evgeni/WebstormProjects/Luxury_Stays_copy/functions/node_modules/@firebase/database/dist/index.standalone.js.map
1020K   /home/evgeni/WebstormProjects/Luxury_Stays_copy/functions/node_modules/@opentelemetry/api/build/esnext
1020K   /home/evgeni/WebstormProjects/Luxury_Stays/functions/node_modules/@firebase/database/dist/index.standalone.js.map
1020K   /home/evgeni/WebstormProjects/Luxury_Stays/functions/node_modules/@opentelemetry/api/build/esnext
1020K   /home/evgeni/.wine/drive_c/windows/system32/winefile.exe
1021M   /home/evgeni/.nvm

┌──(evgeni㉿kali)-[~/ThinkCyber/Project/xe103]
└─$
```